

The Internet of Things, Augmented Reality And Virtual Reality: A FRUITFUL SYNERGY In INDUSTRY 4.0



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This article discusses the concept of fusion reality, where technologies like AR, VR and the IoT converge

Augmented reality (AR) and virtual reality (VR) technologies are rapidly evolving. Improvements to these technologies are, in turn, improving the virtual world user experience, enabling users to do things that have not been possible before.

VR technology allows users to enter the virtual world and interact with virtual things. It is not limited to gaming—other application areas include enhancement of the education system, enablement of doctors to remotely handle surgeries, and contribution to industrial manufacturing processes and quality control, among others.

In addition, AR and VR create opportunities to connect augmented/virtual displays to online platforms for sharing the views of AR/VR users. AR enables viewing virtual information overlaid on the real world.

A company called PTC (Boston, USA) has developed a software application called Vuforia, which provides a platform to combine Internet of Things (IoT) with AR. It allows developers to connect IoT devices with AR applications to augment real objects with a virtual display showing additional useful information, such as real-time data streams.

In the medical field, VR and AR technologies are being implemented to enhance and develop advanced health services. The virtual world is based on enabling technologies that

go beyond the boundaries of the categorised world, in a concept called fusion reality.

Fusion reality is a combination of different virtualisation technologies, such as VR, AR and the IoT. Currently, IoT technologies have transformed many industries in different countries, based on the country's policies. Recently, the German government started a project called Industry 4.0, which aims to improve industrial processes with the help of IoT technologies.

General Electric has a similar project named Industrial Internet. Such projects are expected to boost the revenues of these companies.

Industry 4.0 technologies transforming industrial growth and production

Industry 4.0 refers to a new industrial technology era that is transforming current systems, sensors, machines and workloads. It can help create more efficient processes using the nine pillars of technology: cyber security, AR, big data, autonomous robots, additive manufacturing, simulation, system integration, cloud computing and the IoT (Fig. 2).

Cyber security. Due to demand for better Internet connectivity and communication protocols in Industry 4.0, there is a



Fig. 1: Industry 4.0



Fig. 2: Nine pillars of Industrial 4.0